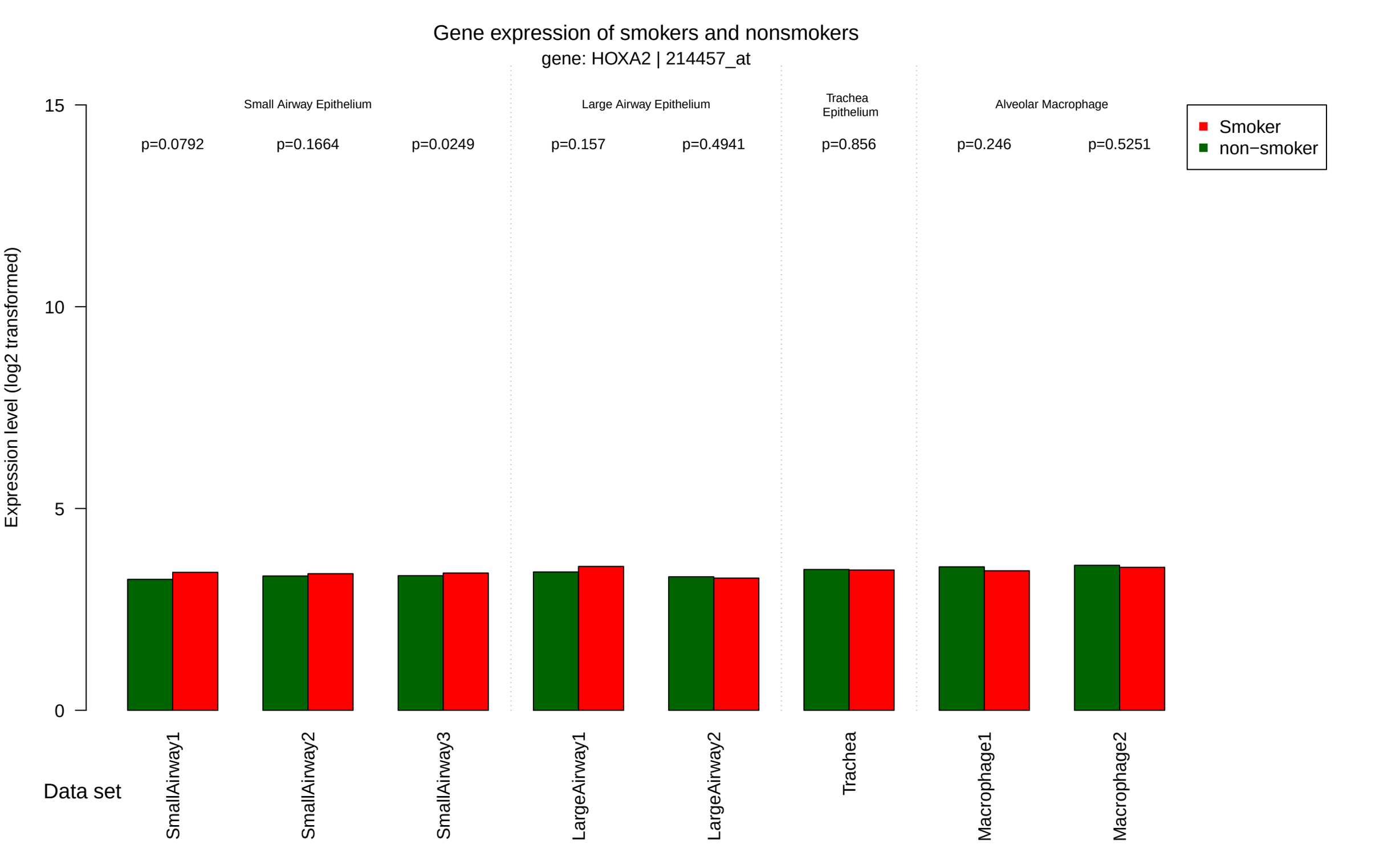
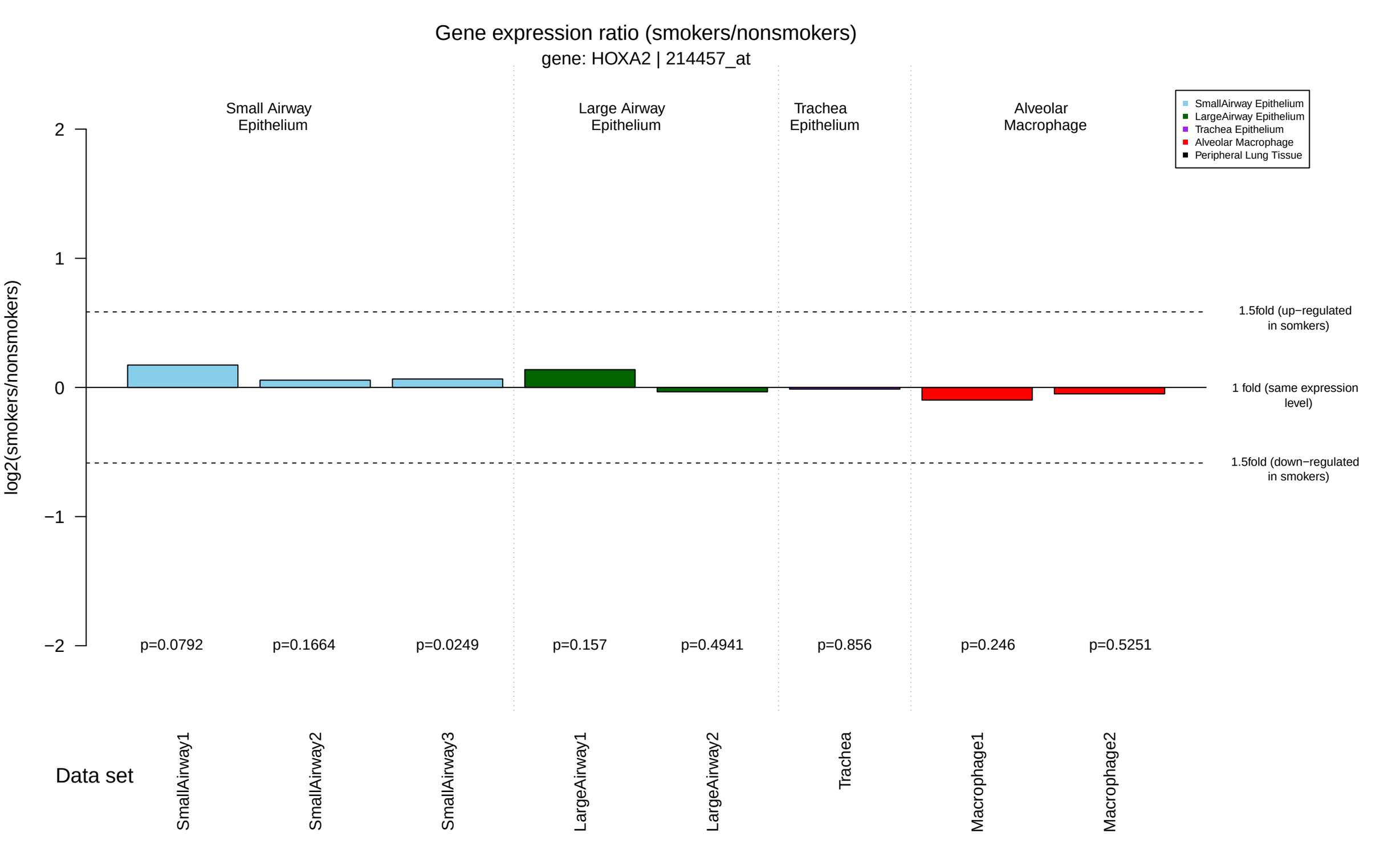
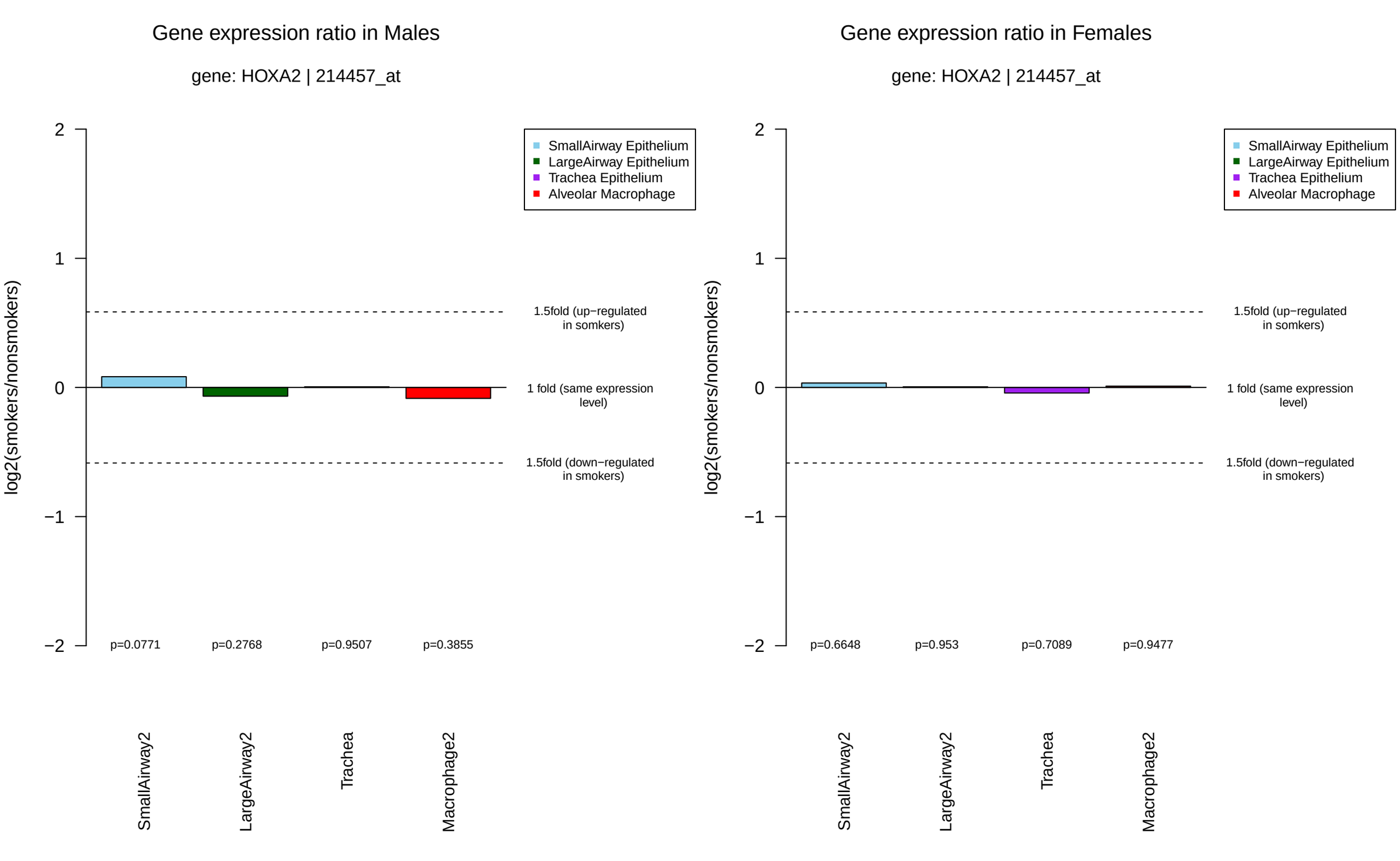
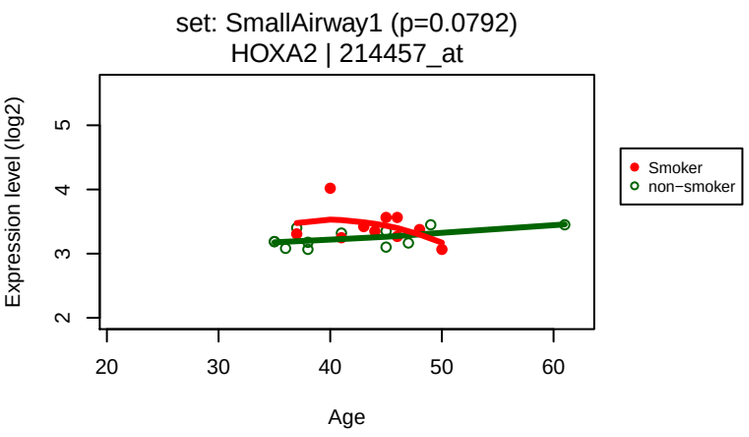
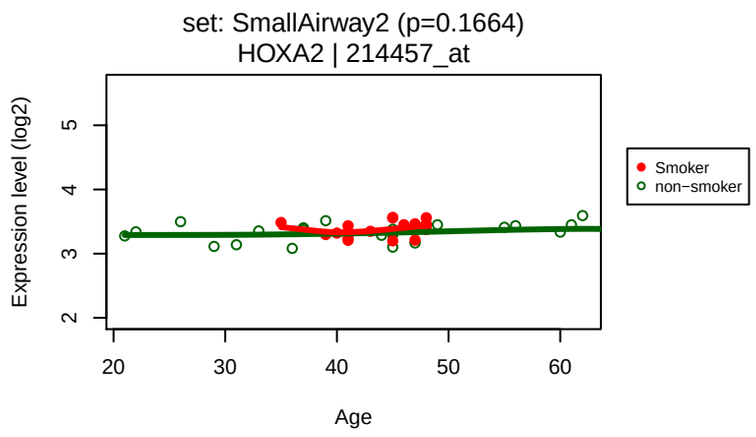
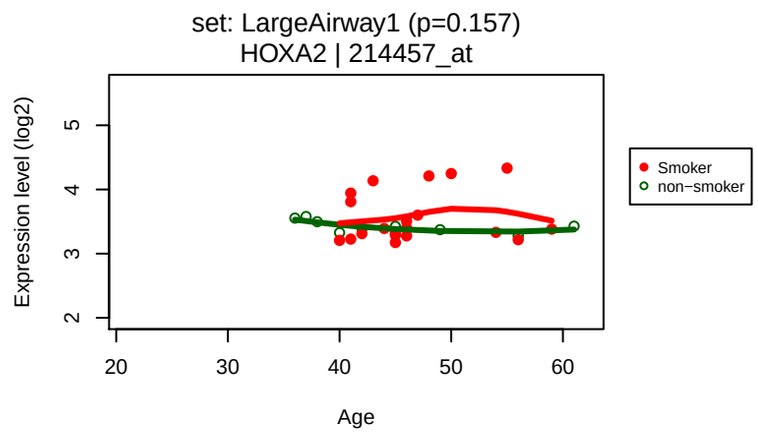

Gene expression of smokers and nonsmokers
gene: HOXA2 | 214457_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0792
p=0.1664
p=0.0249
p=0.157
p=0.4941
p=0.856
p=0.246
p=0.5251
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

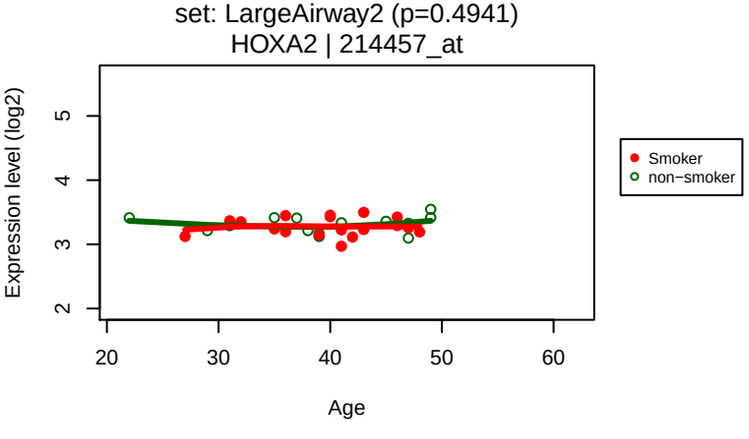

Gene expression ratio (smokers/nonsmokers)
gene: HOXA2 | 214457_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0792
p=0.1664
p=0.0249
p=0.157
p=0.4941
p=0.856
p=0.246
p=0.5251
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

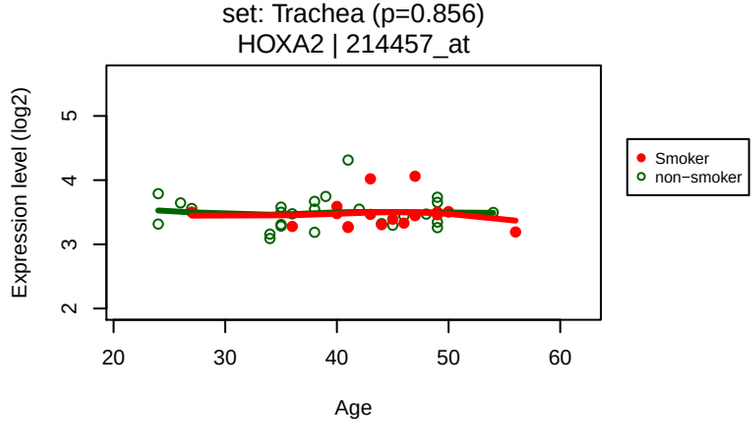

Gene expression ratio in Males
gene: HOXA2 | 214457_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0771
p=0.2768
p=0.9507
p=0.3855
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA2 | 214457_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6648
p=0.953
p=0.7089
p=0.9477
SmallAirway2
LargeAirway2
Trachea
Macrophage2

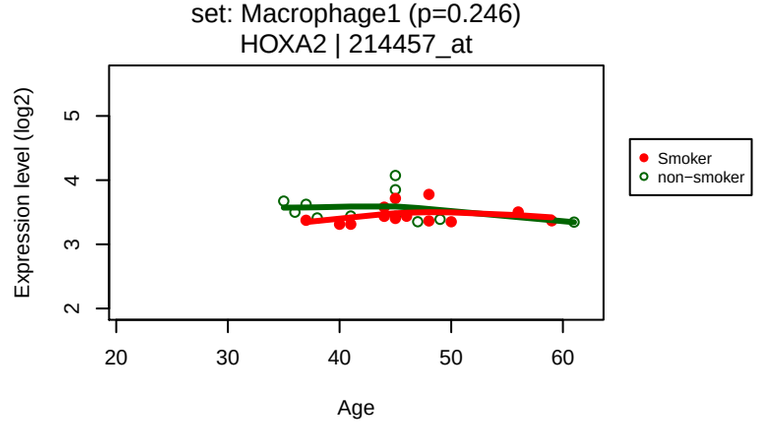

set: SmallAirway1 (p=0.0792)
HOXA2 | 214457_at
Expression level (log2)
Age
Smoker
non−smoker

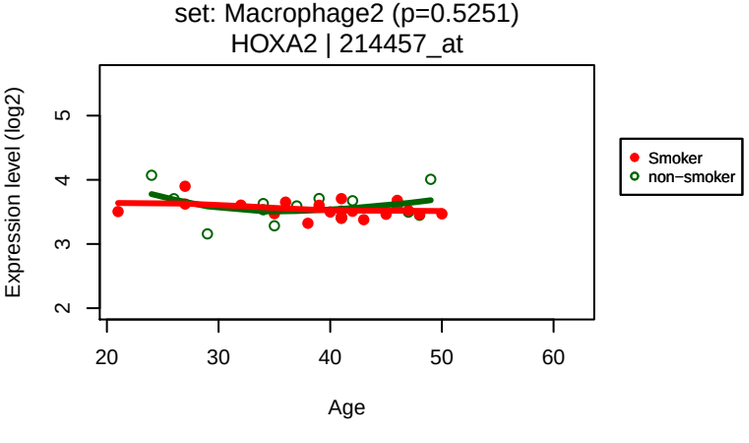

set: SmallAirway2 (p=0.1664)
HOXA2 | 214457_at
Expression level (log2)
Age
Smoker
non−smoker

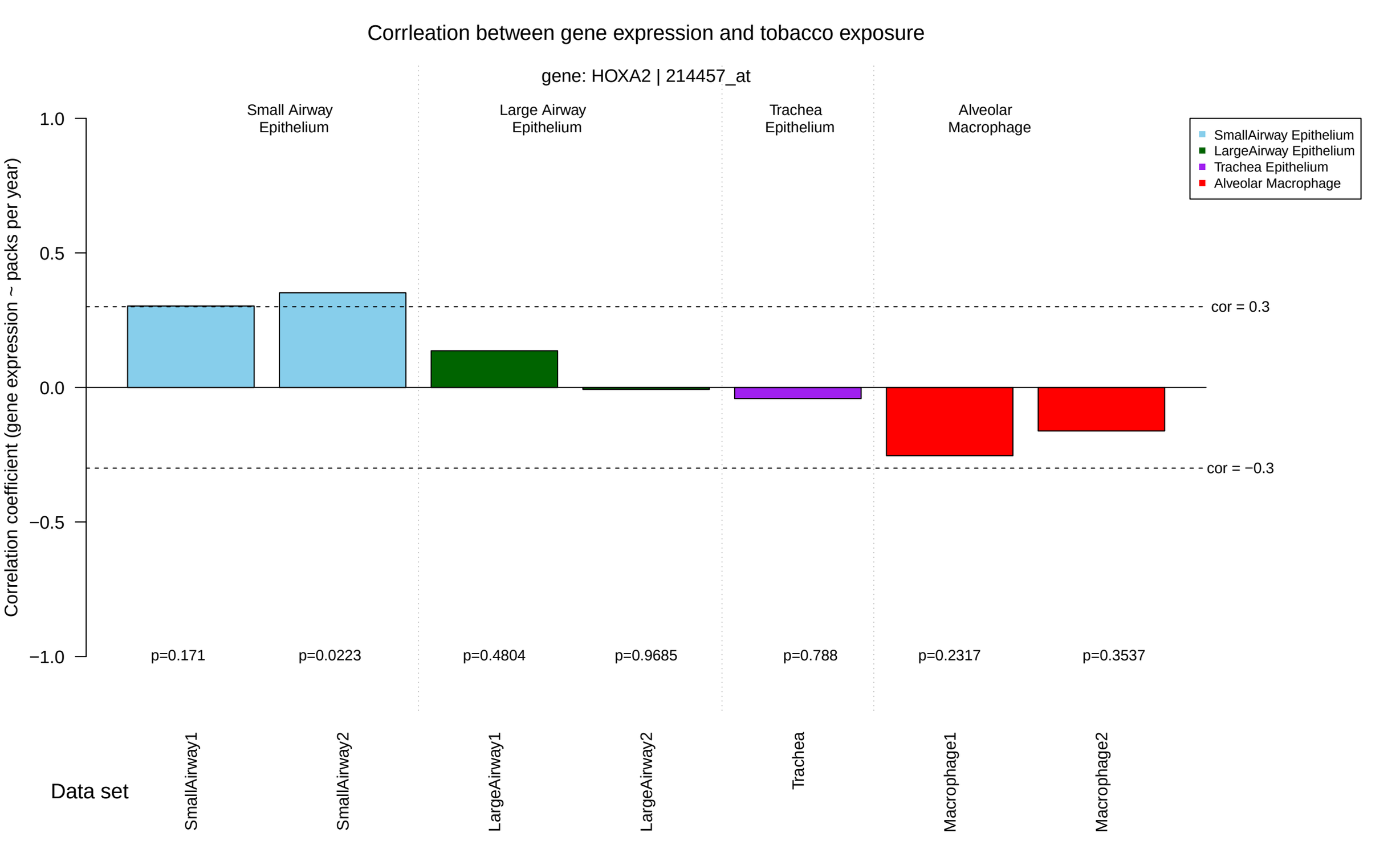

set: LargeAirway1 (p=0.157)
HOXA2 | 214457_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4941)
HOXA2 | 214457_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.856)
HOXA2 | 214457_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.246)
HOXA2 | 214457_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5251)
HOXA2 | 214457_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA2 | 214457_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.171
p=0.0223
p=0.4804
p=0.9685
p=0.788
p=0.2317
p=0.3537
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2